



Bee functional groups respond to vegetation cover and landscape diversity in a Brazilian metropolis

Letícia Vanessa Graf · Israel Schneiberg ·
Rodrigo Barbosa Gonçalves

Received: 3 April 2021 / Accepted: 28 February 2022 / Published online: 12 March 2022
© The Author(s), under exclusive licence to Springer Nature B.V. 2022

Abstract

Context Urbanization is one of the largest causes of habitat loss and fragmentation, factors that modify community structure and can lead to species extinction. Since bees are the main pollinating agents in most ecosystems, losses of bee diversity can have negative consequences on plant diversity and associated services.

Objectives This study aims to understand how bees and their functional traits, such as social behavior, nesting sites, trophic specialization and kleptoparasitism, are affected by an urban landscape.

Methods Ten sites were selected with different levels of urbanization. Bees were sampled with entomological net for 6 h, one day per month, for 10 months.

The response variables evaluated were species diversity, number of bees, and the richness of bee functional traits. Nine landscape metrics were used as predictors, including the proportion of vegetation cover and landscape diversity.

Results Urbanization predictors negatively influenced bee richness but not diversity. Specialist bees were the most susceptible to urbanization, followed by ground nesters and solitary bees. Kleptoparasites also responded negatively to different urbanization metrics. Above-ground nesters and eusocial bees were more resilient to the urban environment and were not affected by urbanization. The proportion of vegetation cover and landscape diversity were the most important predictors for the preservation of bee diversity, followed by the proximity of fragments and the proportion of grasslands.

Conclusions The most responsive and vulnerable functional groups—particularly specialists—should have conservation priority in cities. Analyses of richness and species diversity should not be decoupled from functional traits since doing so may overlook bees' complex response to urbanization.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10980-022-01430-y>.

L. V. Graf (✉) · R. B. Gonçalves
Programa de Pós-Graduação em Ciências Biológicas
(Entomologia), Universidade Federal do Paraná, Caixa
Postal 19020, Curitiba, PR 81531-980, Brazil
e-mail: leticiagraf@yahoo.com.br

R. B. Gonçalves
e-mail: goncalvesrb@gmail.com

I. Schneiberg
Programa de Pós-Graduação em Ecologia e
Conservação, Universidade Federal do Paraná, Curitiba,
PR Mailbox 81531-990, Brazil
e-mail: israelshneiberg@gmail.com

Keywords Atlantic forest · Functional ecology ·
Pollinators · Urban environment

Introduction

It is estimated that, by the end of the century, the world's human population will have increased from 7.6 to 11.2 billion (United Nations 2017). Approximately 54% of people live in cities, which is set to rise to 66% by 2050 (UNRIC 2018). Among the primary ecological consequences of urbanization are habitat loss and fragmentation, processes that can modify the structure of biological communities and, therefore, eventually lead to species extinction (Fahrig 2003; McKinney 2006; Jones and Leather 2012; Seto et al. 2012; Vanbergen et al. 2013) and the disruption of ecosystem services (Tylianakis et al. 2007). Because of the increasing rate of urbanization, there is an urgent need for ecological studies documenting the impact on biological communities as well as the species and functions that are being lost in these environments (Dearborn and Kark 2010). On the other hand, cities that maintain ecological processes can serve as refuges for bees, sustaining pollination services for the urban area and its surroundings (Hall et al. 2017; Banaszak-Cibicka and Żmihorski 2020).

Urbanization results in population losses among the most important groups of insect pollinators (Coleoptera, Diptera, Hymenoptera, and Lepidoptera), which can lead to the local and regional disappearance of numerous species (Jones and Leather 2012; Deguines et al. 2016). Pollinators are essential for maintaining biodiversity since they are responsible for pollinating approximately 90% of angiosperms (Aizen and Feinsinger 2003; Ollerton et al. 2011). With more than 20,500 described species (Ascher and Pickering 2020), bees reflect these changes to a greater degree than other groups because they must feed on floral resources such as pollen and nectar (Ollerton et al. 2011; Cardinal and Danforth 2013). Their ecological and economic importance is estimated at USD 127 to USD 152 billion per year (Bauer and Sue Wing 2016).

In addition to decreasing species richness, urbanization can increase the density of some well-adapted species, causing biotic homogenization (McKinney 2006; Schneiberg et al. 2020). Habitat loss and fragmentation of natural areas affect the proportion of available resources as well as the ability of some species to move through the landscape, and as such are detrimental to bee survival (Fahrig 2003; Niebuhr et al. 2015; Geslin et al. 2016; Boscolo et al. 2017).

Bees have to move through the urban landscape between green areas in order to access the resources they need, which can be more difficult for some species than others (Boscolo et al. 2017). In addition to the availability of natural vegetation, environmental heterogeneity has also been shown to be important for maintaining bee populations (Aguirre-Gutiérrez et al. 2015; Boscolo et al. 2017; Nery et al. 2018).

The life-history traits of bee species influence the way they exploit the landscape and resources around them (Boscolo et al. 2016). For bees, functional traits like social behavior, nesting sites, trophic specialization, and kleptoparasitism have been shown to influence their responses to anthropic changes in the landscape (Banaszak-Cibicka and Żmihorski 2012; Geslin et al. 2016; Silva et al. 2017; Cardoso and Gonçalves 2018; Harrison et al. 2018; Burdine and McCluney 2019; Wenzel et al. 2020; Banaszak-Cibicka and Dylewski 2021). This may result in differences across species in their susceptibility to the impacts of urbanization. For example, Cardoso and Gonçalves (2018) reported a drop of approximately 50% in the richness of solitary bees in an urban area between 1981 and 2015, while the highly eusocial species increased over the same period. Nesting site preference may also affect bees' response to habitat modification. Most bees that nest in the ground appear to be more vulnerable to urbanization than those that nest in wooden cavities or in exposed nests (Geslin et al. 2016).

Some bees depend on a narrow range of host plants, while others supplement their diet with vegetable oils and are considered to be a specialized functional group. Species with high degrees of trophic specialization have shown high sensitivity to anthropic disturbance (Steffan-Dewenter 2003; Winfree et al. 2011; Threlfall et al. 2015; Rocha-Filho et al. 2018; Wenzel et al. 2020). Finally, kleptoparasite bees may be a vulnerable group in disturbed landscapes (Sheffield et al. 2013). The female kleptoparasite lays its egg in the nest of a different bee species, and its larva matures on the provisions stored in the host species' nest (Rozen 2001). Their presence is therefore dependent on their hosts' abundance, meaning that they respond to disturbances at a higher trophic level.

Changes in the taxonomic and functional diversity of bees as a consequence of urbanization should be investigated in different biomes and urban contexts in order to identify effective local and global

conservation strategies (Buchholz and Egerer 2020). Our study seeks to evaluate the response of wild bees to the urban landscape structure in Curitiba, a city in southern Brazil. We began by identifying variations in the species composition across study sites, following which we modeled nine landscape metrics. We expect that environmental heterogeneity, vegetation cover, and isolation are the most important explanatory variables for bee richness and diversity, which will disproportionately affect solitary, below-ground nesters, and specialized bees. Below-ground nesters are likely to be sensitive to the proportion of grassland cover, while above-ground nesters will depend on tree cover. Kleptoparasites are likely to be among the most affected by changes in landscape variables.

Material and methods

Study area

The study was carried out in the city of Curitiba, the adjacent municipality of São José dos Pinhais, both of which are located in the state of Paraná in southern Brazil (Fig. 1A–B). The Curitiba metropolitan region currently comprises 29 municipalities with a population of 3,615,027 inhabitants (IBGE 2020). The city of Curitiba has 1,933,105 inhabitants, with a population density of 4027 people per km², while São José dos Pinhais has 329,058 inhabitants and a population density of 279.16 people per km² (IBGE 2020). The climate in the region is subtropical (average summer temperature of 22 °C, average winter temperature of 17 °C, average rainfall of 1480 mm; INMET—Instituto Nacional de Meteorologia). The local plant cover is a mix of Araucaria forests and natural open fields within the Atlantic Forest ecoregion.

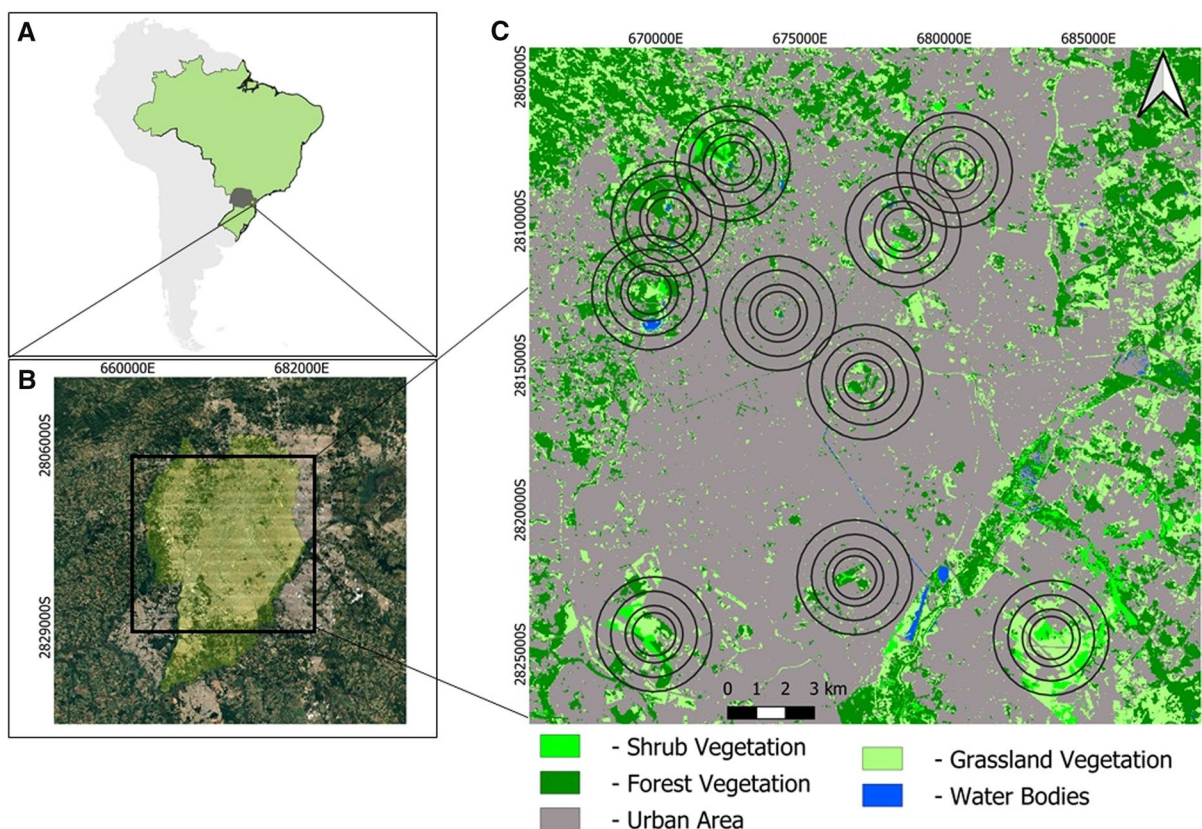


Fig. 1 The city of Curitiba, southern Brazil, and the ten study sites with analyzed buffers. **A** Brazil (green) and state of Paraná (gray). **B** Curitiba (green), **C** study sites and buffers (750 m, 1 km, 1.5 km, and 2 km)

We selected ten sampling sites (Fig. 1C, Online Resource 1), nine in Curitiba and one in São José dos Pinhais. Three sampling sites had already been studied for temporal responses of bees to urbanization in previous studies: Site 1 (Taura and Laroca 2001), Site 3 (Cardoso and Gonçalves 2018), and Site 10 (Martins et al. 2013). The other seven sites are located in public parks and military areas. No private areas were selected due to the risk of private green space being built over during the study period. We chose the sites using the following criteria: different proportions of vegetation around the central collection point (centroid) in a radius of 2 km; presence of herbaceous-shrub vegetation with flowering plants; a minimum distance of 1 km between fragment centroids to ensure sampling independence between sites (Ferreira et al. 2015; Geslin et al. 2016); the ability for collectors to carry out survey activities safely.

Bee sampling

Sampling was carried out one day per month for ten months between August 2017 and May 2018, for a total of ten days of sampling at each site. The sampling period was designed so as to exclude the winter months of June and July when there is minimal bee activity (average temperature of 9.3 °C). Sampling was carried out for six hours per day (9:00 a.m. 03:30 p.m., with a half-hour break) for a total of 60 h per site. No surveys were carried out on cloudy or rainy days due to reduced bee activity. Flower-visiting bees were collected with entomological nets by one collector (LVG). The surveys were carried out within a radius of up to 200 m around the centroid and aimed to visit every flowering plant in the study area during the sampling period. This distance was defined to standardize the sampling area regardless of fragment size, as well as to be a manageable size for a single day of sampling.

We chose to focus on wild bees only, meaning that individuals of the Western honey bee (*Apis mellifera*) were not collected as including them would jeopardize the sampling and processing of the other specimens. All other specimens collected were prepared for identification on entomological pins, dried at temperatures between 40 and 60 °C for three days, labeled, identified, recorded in a database, and finally deposited in the Jesus Santiago Moure Entomological Collection (DZUP) at the Federal University of

Paraná. Bee classification followed Melo and Gonçalves (2005), where the whole group is treated as a single family (Hymenoptera, Apoidea, Apidae sensu lato). Genera were identified using Silveira et al. (2002), following which species-level identifications were made considering taxonomic revisions (see a summary in the catalog of Moure et al. 2012) and comparison with DZUP specimens.

Functional groups were identified using the classification of Martins et al. (2013) and biological information summarized in Michener (2007). The data were reviewed by the expert consultants listed in the acknowledgments section. We used three main categories: behavior, nesting preference, and trophic specialization. For behavior, bees were separated into solitary and eusocial, with the latter group further divided into primitive (totipotent reproductive caste) and highly eusocial (fixed reproductive caste) groups to encompass variation in reproductive behavior. Bees were separated according to nesting preferences between those nesting in the soil (ground-nesters) and those that nest above ground (exposed nests, cavities, and other substrates). With respect to trophic specialization, species were separated into specialists (oligolectic and oil bees) and generalists. Due to its particularities, the kleptoparasite group was analyzed separately from the other categories.

Landscape data

We used a Landsat 8 image of the study area, which has a 30 m resolution. The acquisition date was April 25th, 2016 (ID: LC08_L1TP_220078_20160425_20170326_01_T1; path: 220; row: 78). The georeferencing image underwent geometric correction and atmospheric correction (DOS1) before the classification procedure. Image classification was performed using the Semi-Automatic Classification Plugin (SCP) (Congedo 2016) with QGIS version 3.10.0 (QGIS Development Team 2019). The resulting land-use map was manually post-processed to correct the minor errors of the semi-automatic classification. We considered five units, defined as follows: urban area, water bodies, forest vegetation, shrub vegetation, and grassland vegetation. The grassland unit includes natural open fields, managed grass, exposed soil, and areas covered with ruderal plants. In this study, four buffers were used to obtain the spatial metrics (750 m, 1 km, 1.5 km, and 2 km) (Fig. 1), defined

based on the flight area of most bees (Araújo et al. 2004; Ferreira et al. 2015; Geslin et al. 2016) because flight area and other processes are scale-dependent (Jackson and Fahrig 2015; Miguet et al. 2016). We also considered a radius of 500 m but did not use it because most study sites were surrounded by a green area that occupied more than 50% of this buffer, impairing the analyses.

We calculated nine landscape metrics based on the five units described above, as follows: Shannon's diversity index of the landscape, proportion of grass cover, proportion of arboreal cover (sum of shrub and forest cover), proportion of vegetation cover (sum of grass, shrub, and forest cover), largest patch index, edge density, number of patches, mean Euclidean nearest neighbor distance (which measures the patch isolation), and mean shape index. The mean shape index is the simplest way to measure the complexity of the class shapes of a landscape (Flores et al. 2019). Shannon's diversity index of the landscape was used to describe landscape diversity (environmental heterogeneity), considering the number and proportion of landscape classes (Boscolo et al. 2017; Nery et al. 2018). Its value equals minus the sum of the total landscape proportion of each land cover unit multiplied by its natural logarithm. This metric increases with growing numbers of land cover units and/or when the proportional area of these units becomes more equitable. These metrics reflect the landscape composition and configuration and can be expected to affect our ecological variables, with specific expectations related to the different trait groups. All metrics were calculated using Fragstats v.4 (McGarigal et al. 2012).

Data analysis

Bee diversity, bee richness, and richness of each functional group were selected as the response variables of interest to explore how bee diversity changes along variables reflecting increasing urbanization. We used the Shannon index as a diversity index and complemented the bee diversity analysis with a dispersion diagram between species richness and Pielou equability index (J') (Pielou 1966) for each study site to understand the effect of the presence of eusocial bees on the result (Melo 2008) (Online Resource 2). To describe the differences in species composition across study sites, the value of beta diversity was obtained

using the Bray–Curtis dissimilarity index, which considers species abundance (values close to 0 indicate greater similarity in species composition). Beta diversity was calculated using the *betapart* package (Baselga and Orme 2012), which allows values to be separated into two components: the replacement of species (turnover) and the loss of species (nestedness) across sites (Baselga 2010).

To select the spatial extent at which each landscape metric best predicted diversity, total richness, and functional group, we performed univariate fits for each metric and scale and selected the one with the highest coefficient of determination (R^2) (Boscolo and Metzger 2009; Moreira et al. 2015; Miguet et al. 2016; Schneiberg et al. 2020). The proportion of vegetation cover and the landscape diversity showed high values of R^2 for most of the biological variables evaluated, particularly at the smallest buffer analyzed (750 m) (Online Resource 3). Since the correlation between these two variables at this scale was high (0.88), and both were considered important, the selection of models was carried out separately for each one with the remaining predictors.

Before performing the model selection analysis, we checked for multicollinearity between the predictor variables using Pearson correlation. We removed the variables with strong collinearity (i.e., more than 0.7) with the proportion of vegetation cover or landscape diversity (performed separately for each one), which were the variables with the highest R^2 values concerning most of the biological variables. In the event that a pair of variables had strong collinearity (for example, between isolation and mean shape index), we chose the one that had the highest R^2 .

All landscape metrics included in the model selection were centered to zero using the *standardize* function in the *vegan* package due to the range of absolute values that varied naturally. In order to access the complexity of the relationships between landscape structure, bee species, and functional diversity, we compared the multiple competing hypotheses using model selection analysis and a selection criterion corrected for small sample sizes (Akaike Information Criterion—AICc). The analysis also provides the difference between the AICc value of a given model and the most parsimonious model (ΔAICc), as well as the weight of evidence (w_i) of each model. Models with $\Delta\text{AICc} < 2$ were considered equally plausible, and no model was considered when the no-effect

model showed $\Delta AICc < 2$. Models with a high weight of evidence have a high probability of being plausible explanations for the selected response variable. The model selection analysis was carried out using the *AICcmodavg* package (Mazerolle 2019). A Poisson distribution was used for count data (richness) and Gaussian distribution for species diversity. All statistical analyses were performed in R version 3.6.2, with R Studio version 1.2.5033 (R Development Core Team 2016) using the *vegan* package (Oksanen et al. 2013).

When analyzing the different proportions of vegetation cover and Shannon's diversity index of each sampling site (Fig. 1 and Online Resources 9), Site 1 clearly stood out for its lower values compared to all the other sites. For this reason, the selection of models was also performed without this site to explore the relationships between the response variables and the predictors for the data from the remaining sites.

Results

A total of 8938 specimens were sampled, belonging to 288 species and the five subfamilies that occur in Brazil. The abundance of each species and the sites where they were collected can be found in Online Resource 4. The overall value of beta diversity across all study sites was 0.78, with a turnover value of 0.76 and nestedness of 0.02. Of the species collected, 11.8% were kleptoparasites (not included in the other functional groups), 68.8% were solitary, and 19.4% were eusocial (5.6% highly eusocial and 13.9% primitively eusocial). Meanwhile, 55.6% nested in the soil and 32.7% nested above ground, while 20.5% were specialists and 67.7% generalists. The species and their functional groups are shown in Online Resource 5. The values of species richness overall and within each functional group, as well as Shannon's diversity index for each area, are found in Online Resource 6.

The GLM output resulting in plausible models ($\Delta AICc < 2$) is shown in Table 1 alongside the respective vegetation cover and in Table 2 with landscape diversity. Three models described the variation in total bee richness and showed that bee species richness increased with vegetation cover and landscape diversity. The isolation of the fragments was also considered important for total bee richness, but only in addition to vegetation cover (Tables 1, 2, Fig. 2A–B).

This relationship between vegetation cover and landscape diversity indicated the strongest response among specialist species (assessed through the intercept values), followed by the below-ground nesters and solitary bees, with generalist and kleptoparasite bees the least affected (Tables 1, 2, Fig. 2C–N).

The isolation of the fragments alone was only relevant for the generalist and kleptoparasite groups (Fig. 2K, N). The grassland cover proportion, combined with landscape diversity, was important for below-ground nesters and specialists. Contrary to predictions, landscape variables had no effect ($\Delta AICc > 2$ for all models) on bee diversity, above-ground nesters, or either group of eusocial bees (primitively and highly eusocial) (Online Resources 7, Online Resource 8). The results of the GLM performed without Site 1 showed that the relationship of the functional groups that showed the highest vulnerability to urban landscapes with vegetation cover and with landscape diversity is maintained, namely below-ground nesters ($w_i = 0.40$ and 0.57 , respectively; $\alpha = 0.09$ and 0.08 , respectively) and specialist bees ($w_i = 0.61$ and 0.60 , respectively; $\alpha = 0.18$ and 0.17 , respectively).

Discussion

Only about half of the bee species (66) were found in the most urbanized sampling unit (Site 1), which had 7% vegetation cover. This is a considerable loss of species compared to the remaining sites, with 111 to 134 bee species (Online Resource 6). These nine other study sites had a vegetation cover of 33% within a 750 m radius. Our result is consistent with findings in the literature, where a drop-off in bee richness is observed at a threshold of around 35 to 40% vegetation cover (Ferreira et al. 2015; Boscolo et al. 2017; Wenzel et al. 2020). Even though there is no data between the 7 and 33%, this consistency across different locations indicates that this threshold value may be useful in determining conservation strategies based on the proportion of green areas in urban landscapes.

Even though we did not sample sites with intermediate vegetation cover (between 7 and 33%), the selected models show that the values of Site 1 are lined up with the values of the other sites (Fig. 2A–N). Also, the results for the main vulnerable groups

Table 1 Model selection for bee responses with the vegetation cover proportion among the predictors

Response variable/Predictor variable	Δ AICc	Weight (w_i)	Intercept (α)	
Bee richness				
Proportion of vegetation cover	0.00	0.63	0.15	
Proportion of vegetation cover + Isolation	1.73	0.27	0.11	−0.07
Isolation	4.14	0.08	−	
No effect model	6.52	0.02	−	
Solitary bee richness				
Proportion of vegetation cover	0.00	0.70	0.18	
Proportion of vegetation cover + Isolation	2.37	0.21	−	
Isolation	5.39	0.05	−	
No effect model	5.71	0.04	−	
Below ground nester bee richness				
Proportion of vegetation cover	0.00	0.73	0.22	
Proportion of vegetation cover + Isolation	2.23	0.24	−	
Isolation	7.05	0.02	−	
No effect model	9.81	0.01	−	
Specialist bee richness				
Proportion of vegetation cover	0.00	0.87	0.36	
Proportion of vegetation cover + Isolation	3.97	0.12	−	
Isolation	9.93	0.01	−	
No effect model	10.76	0.00	−	
Generalist bee richness				
Isolation	0.00	0.48	−0.11	
Proportion of vegetation cover	0.51	0.37	0.10	
Proportion of vegetation cover + Isolation	2.85	0.12	−	
No effect model	5.60	0.03	−	
Cleptoparasitic bee richness				
Proportion of vegetation cover	0.00	0.44	0.29	
Isolation	0.30	0.38	−0.30	
No effect model	3.25	0.09	−	

We calculated the parameters of relative difference (Δ AICc), relative likelihood (w_i), and the intercept of models. All variables were selected using a buffer of 750 m around the centroid. The plausible models are in bold

(below-ground nesters and specialist bees) are the same for the nine remaining sites. This shows that, even though the values of the predictors are much lower at Site 1 than the others, this site should not be considered an outlier.

The overall beta diversity value of turnover among the study sites was high, showing that high species replacement was occurring, meaning that the bee community changed considerably around the city. On the other hand, total bee species loss (nestedness), a richness metric, had very low importance for the overall beta diversity. Such finding suggests that richness is only one criterion for conservation strategies, as each species and functional group has particular survival needs. This result is congruent with the selection of models performed, demonstrating

that the overall effect of urbanization on bee populations is different from the effect on specific functional groups.

Through the selection of models, we found that bee richness, but not diversity, is associated with vegetation cover and landscape diversity in Curitiba, contrary to expectations. Examining diversity, we speculate that bee diversity in our system could be influenced by the high abundance of a few highly eusocial stingless bees (Online Resource 2). A problem with diversity indices is that they encompass values of richness and equability, which can mask the responses of individual species (Melo 2008; Wilson and Jamieson 2019). Many studies that focus on the magnitude of the effect of landscape structure on species diversity and richness have

Table 2 Model selection for bee responses with landscape diversity among the predictors

Response variable/Predictor variable	Δ AICc	Weight (w_i)	Intercept (α)	
Bee richness				
Landscape diversity	0.00	0.86	0.18	
Landscape diversity + Proportion of grassland	3.63	0.14	–	
No effect model	13.60	0.00	–	
Solitary bee richness				
Landscape diversity	0.00	0.76	0.23	
Landscape diversity + Proportion of grassland	2.31	0.24	–	
No effect model	11.85	0.00	–	
Below ground nester bee richness				
Landscape diversity	0.00	0.64	0.26	
Landscape diversity + Proportion of grassland	1.15	0.36	0.21	0.08
Proportion of grassland	12.14	0.00	–	
No effect model	12.92	0.00	–	
Specialist bee richness				
Landscape diversity	0.00	0.62	0.42	
Landscape diversity + Proportion of grassland	1.96	0.23	0.35	0.13
Landscape diversity + Isolation	3.57	0.10	–	
Landscape diversity + Proportion of grass + Isolation	7.28	0.02	–	
Proportion of grass + Isolation	8.33	0.01	–	
Proportion of grassland	8.49	0.01	–	
Isolation	10.34	0.00	–	
No effect model	11.17	0.00	–	
Generalist bee richness				
Landscape diversity	0.00	0.77	0.13	
Landscape diversity + Edge density	4.14	0.10	–	
Landscape diversity + Proportion of grassland	4.28	0.09	–	
Edge density	7.20	0.02	–	
Proportion of grassland + Edge density	8.36	0.01	–	
Proportion of grassland	9.50	0.01	–	
Landscape diversity + Proportion of grassland + Edge density	10.08	0.00	–	
No effect model	10.14	0.00	–	
Cleptoparasitic bee richness				
Landscape diversity	0.00	0.67	0.35	
Proportion of grassland	3.13	0.14	–	
No effect model	3.72	0.10	–	

We calculated the parameters of relative difference (Δ AICc), relative likelihood (w_i), and the intercept of models. All variables were selected using a buffer of 750 m around the centroid. The plausible models are in bold

assumed that all species are functionally equivalent, but bee species have different social organizations, nesting, and plant preferences (Michener 2007), and these functional traits lead to different responses to urbanization (Grime 1998; Hubbell 2001; Cariveau and Winfree 2015; Martins et al. 2017).

In Curitiba, almost 70% of the sampled species were solitary, as expected (Danforth et al. 2019). Unlike social species, solitary bees were positively associated with vegetation cover and landscape diversity. This response is in line with solitary bees' higher susceptibility to the effects of urbanization

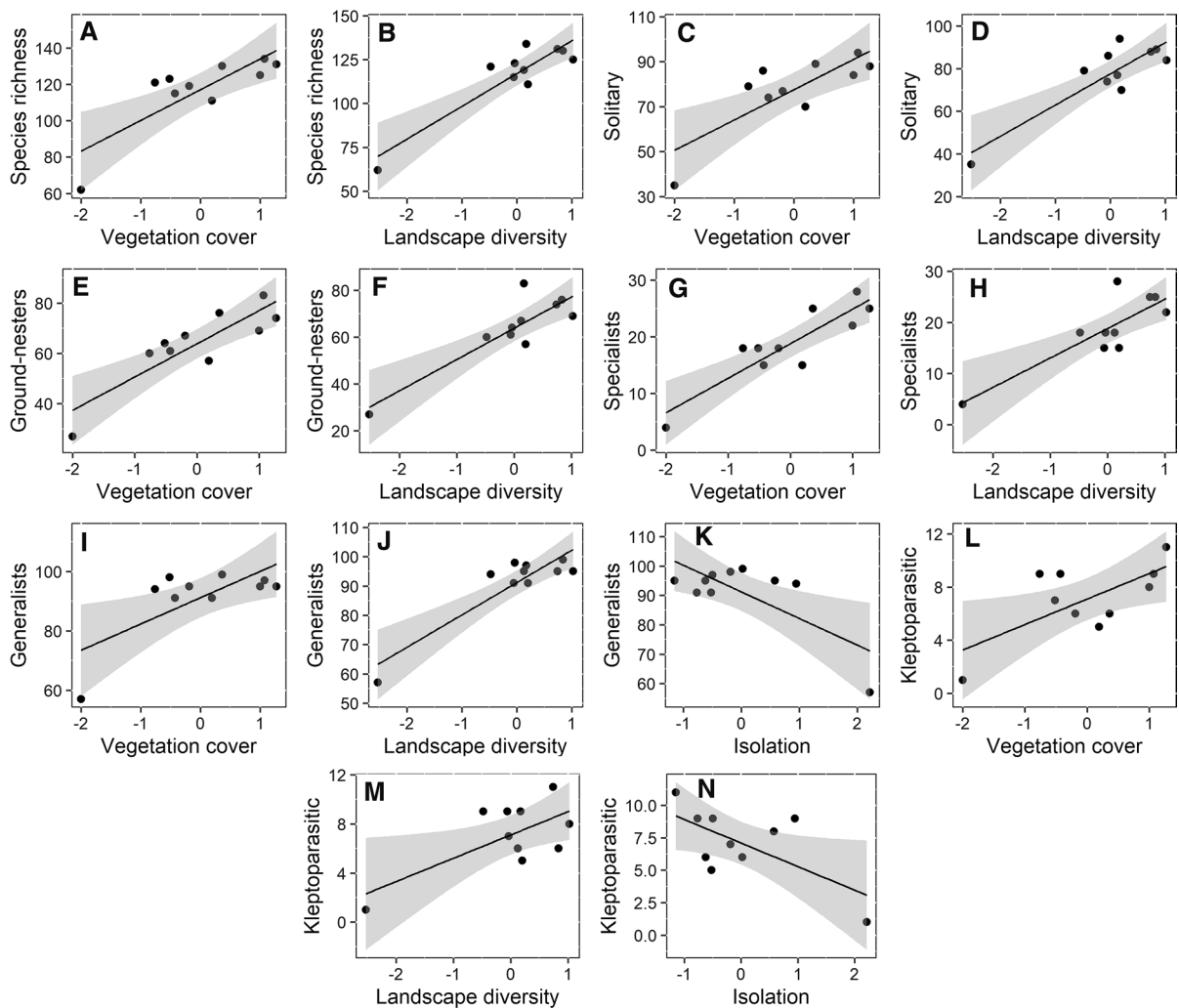


Fig. 2 Distribution of bees and functional trait richness along the gradient of vegetation cover, landscape diversity, and isolation in the city of Curitiba, southern Brazil. **A** Bee richness $\times$ vegetation cover. **B** Bee richness $\times$ landscape diversity. **C** Solitary $\times$ vegetation cover. **D** Solitary $\times$ landscape diversity. **E** Ground nesters $\times$ vegetation cover. **F** Ground nesters $\times$ land-

scape diversity. **G** Specialists $\times$ vegetation cover. **H** Specialists $\times$ landscape diversity. **I** Generalists $\times$ vegetation cover. **J** Generalists $\times$ landscape diversity. **K** Generalists $\times$ isolation. **L** Kleptoparasites $\times$ vegetation cover. **M** Kleptoparasites $\times$ landscape diversity. **N** Kleptoparasites $\times$ isolation

than eusocial bees, probably due to their lower abundance (Chapman and Bourke 2001), ephemeral foraging period, lower behavioral and ecological flexibility (Banaszak-Cibicka and Zmihorski 2012), and more frequent host plant specialization.

Stingless bees are an important component of the eusocial functional group in tropical areas. Their perennial colonies with stored food can be valuable in disturbed habitats with scarce resources, such as cities (Zanette et al. 2005). Usually, the most abundant

species in the Brazilian cities belong to this tribe. *Trigona spinipes* (Fabricius, 1793) is one of the most resilient species in Curitiba (Cardoso and Gonçalves 2018), as well as in other cities around the country (Zanette et al. 2005; Jaffè et al. 2016). We also found a high abundance of the small-sized *Plebeia* Schwarz and *Tetragonisca* Moure species. Because the sampled stingless bees are small (< 1 cm) and therefore fly short distances, they may be expected to be more sensitive to urbanized landscapes, responding

to smaller-scale changes (Steffan-Dewenter et al. 2002). However, this was not the case here, indicating that bee size alone may not reflect the overall sensitivity. In contrast, there are some large species of Meliponini, like species of *Melipona* Illiger, which can be vulnerable to urbanization (Ferreira et al. 2015)—one example is *M. bicolor* Lepeletier, 1836. These species need large tree hollows to nest (Silva 2008), which are not usually found in urban sites. Their nests are also not found in urban cavities such as walls.

Although eusocial bees comprise less than 20% of all species, they are more abundant. Neither the fixed caste nor the totipotent eusocial subgroups responded to urbanization. These groups include totipotent bumblebees, small carpenter bees, and sweat bees, which have been shown to be resilient, with high abundance even in highly urbanized areas. This indicates that the presence of a colony, even ephemeral, and associated longer foraging periods may be related to lower sensitivity (Michener 2007). While social species occur throughout the urban landscape without significantly decreasing their richness (Oliveira et al. 2009; Aguirre-Gutiérrez et al. 2015; Ferreira et al. 2015; Silva et al. 2017; Cardoso and Gonçalves 2018), this pattern drastically changes in agricultural sites, where they are much more sensitive to changes than solitary bees. Researchers have found that social species respond more strongly to patch isolation, probably due to correlated differences in nesting substrate, and responded more to pesticides, reflecting colony bioaccumulation and their greater and longer exposure periods to toxic substances (Williams et al. 2010).

Although bees can nest in a myriad of substrates, they can be broadly classified into species that nest in the soil and those that explore different kinds of above-ground surfaces. This distinction is important because most species nesting in the soil need permeable areas, and their higher response to urbanization is commonly related to high amounts of impermeable areas within the cities (Geslin et al. 2016; Pereira et al. 2020; Wenzel et al. 2020). Even with permeable areas, urban green areas usually have extensive surfaces with dry and compacted soil, offering few opportunities for bee nesting (Xie et al. 2013; Rocha-Filho et al. 2018). On the other hand, above-ground nesters can benefit from urban forest sites and human-made cavities, though the above-ground functional group did not respond to the landscape predictors in

the present study. Harrison et al. (2018) observed a negative response to urbanization, but other studies showed that cavity-nesting bees responded positively to the urban landscape, possibly because many species can exploit human constructions as a substrate (Zanette et al. 2005; Wenzel et al. 2020). These conflicting results may be related to bee responses to different attributes of cities, such as the degree of urbanization, biome, and local species pool (Rocha-Filho et al. 2018; Buchholz and Egerer 2020).

Bees with specialized host plant relations are the most vulnerable to urbanization, according to our results. These bees are more vulnerable to plant population losses (Packer et al. 2005), especially in urbanized areas (Cariveau and Winfree 2015). Plant losses and homogenization can suppress species that provide pollen and oil for specialist bees. Thus, besides needing a specific food supply, this group has other biological attributes that respond negatively to urbanization. Notably, all the specialist species we sampled are solitary, and 92% build subterranean nests. This functional conjunction explains why this trait is considered the most vulnerable to urbanization, with greater chances of suffering local extinctions (Biesmeijer et al. 2006; Harrison et al. 2018). One example is the oligolectic bees belonging to the tribe Neopasiphaeini (Almeida et al. 2019), which have declined in Curitiba according to the available historical data (Taura and Laroca 2001; Martins et al. 2013; Pereira et al. 2020) and are extinct in the more urbanized sites. The Neopasiphaeini species usually have only one generation per year, with adults foraging only for a few weeks, another trend that can contribute to their vulnerability.

Therefore, as expected, all the results presented here demonstrate that plant specialist bees, ground nesters, and solitary species are the ones that contribute most to the lower total bee species richness in more urbanized areas. We also found that generalist species responded positively to vegetation and landscape diversity. Examining the distributions in Fig. 2, generalist richness (Fig. 2I, J) has a trend similar to overall bee richness, with a strong contribution of Site 1 (Fig. 2A, B). Indeed, the response of generalist species disappeared when this site was removed from the analysis, indicating that this group is less sensitive to the effects of urbanization.

Kleptoparasites had greater rates of dispersion across sites (Fig. 2L–N). These bees were almost

absent at the most urbanized site, as previously reported by Taura and Laroca (2001). Cuckoo bees depend on the presence of their host and the resources it provides (Rozen 2001; Michener 2007), meaning that they are directly affected by landscape disturbances. Thus, a linear response to environmental metrics can be attenuated in some circumstances. A pair of long-term studies by Martins et al. (2013) and Cardoso and Gonçalves (2018) monitored Sites 10 and 3, respectively, and did not find evidence of declines in kleptoparasites. As discussed by Sheffield et al. (2013), kleptoparasites can be indicative of the status of the total bee assemblage and, as expected, they were more affected by the metrics evaluated than the bee community at large (assumed by the higher intercept values), indicating that monitoring in urban areas should also focus on these bees.

Biotic homogenization encompasses the processes by which a community loses its biological distinctions at any level of organization, whether genetic, taxonomic, or functional. Functional biotic homogenization is related to greater similarity in species' roles in a community due to the loss of unique traits (Olden and Rooney 2006). Many studies describe the functional biotic homogenization caused by urbanization in terms of the loss of trophic specialist species (McKinney 2006; Deguines et al. 2016; Schneiberg et al. 2020). However, solitary and ground-nesting groups also declined in urbanized sites, providing a new perspective on how biotic homogenization should be investigated for the group.

Among the nine landscape predictors tested, the proportion of vegetation cover and Shannon's landscape diversity index had the greatest explanatory power. The former represents the natural cover of habitat in relation to the total area of the landscape and is usually the most common explanatory variable for urban species richness across numerous taxa, including bees (Fahrig 2003; Calvillo et al. 2010). Some studies have focused on the proportion of impervious surfaces in urban areas as it is assumed to negatively correlate with habitat area (e.g., Geslin et al. 2016). Here, we used the proportion of vegetation cover, further broken down into different units (grass, shrub, and forest cover) to analyze different responses. The second predictor is a measure of environmental heterogeneity, an important predictor noted by some authors (Fahrig et al. 2011; Moreira et al. 2015). Bees can benefit from heterogeneous landscapes due

to increased food and nesting resources (Aguirre-Gutiérrez et al. 2015; Boscolo et al. 2017; Nery et al. 2018), as do other pollinators, particularly butterflies (Orlandin and Carneiro 2021).

When analyzing vegetation cover in different classes, we found that only grassland cover was significant for specialists and ground-nesters. Grasslands provide complementary resources for bees, such as floral and nest-building materials and nesting substrates (Nery et al. 2018). Open vegetation formations are the most vulnerable areas to urbanization (Boscolo et al. 2017; Nery et al. 2018), frequently neglected by conservation plans compared to forested sites. Although some bees are dependent on forest vegetation, especially some cavity nesters, we did not find a significant response for any of the functional traits evaluated to this vegetation class. For example, birds have been shown to increase their richness with more forested areas in urban environments (Schneiberg et al. 2020). This highlights the different needs of distinct organisms (Boscolo et al. 2016), which can require a variety of conservation approaches. Another important prediction was the proximity of the natural fragments, corroborating the importance of functional connectivity for bee survival (Boscolo et al. 2017).

The proportion of grassland cover was highly correlated with the proportion of vegetation cover, which has been removed from the analysis performed with this metric. However, it showed a low correlation with landscape diversity, which proved to be an important metric for the main vulnerable groups in the urban environment (below-ground nesters and specialist bees). This highlights the importance of performing the analysis with the metrics of landscape diversity and vegetation cover separately. Among the other metrics assessed here, only patch isolation proved to influence the bee richness of some groups, although as a secondary metric.

A recent review on the importance of multi-scale studies underscored the ability of some factors to impair analysis (Miguet et al. 2016), especially the decreasing of variance values as the scale increases. We did not find a relevant decrease from 750 to 2 km, and the landscape diversity variance increased instead (Online Resource 9). The smaller buffer (750 m) had the greatest degree of explanatory power for our data, suggesting that nearby resources and nesting substrate are important, both in Curitiba and other cities

(Ferreira et al. 2015; Geslin et al. 2016). Although the maximum flight capacity of many species is greater than the buffer used, bees forage close to their nest, which reduces the landscape scale they respond to (Araújo et al. 2004; Wright et al. 2015; Geslin et al. 2016).

With regard to the satellite image used in this study, we chose Landsat 8, which has a 30 m resolution, because it captures a good estimation of class-level metrics (e.g., vegetation cover, urban areas), it has multitemporal image availability and free data acquisition. Although Landsat satellites provide medium-resolution imagery, it has been used in many bee-based studies (Machado et al. 2020; Ferreira et al. 2020; Coutinho et al. 2021), which makes comparison between studies less biased. However, high resolution images can improve the precision of the data and also provide more detailed landscape perspective of patch-level metrics by highlighting corridors and potential stepping stones. In this way, better resolution images like Sentinel (10 m) could be used for this type of studies in the future to obtain more accurate data.

Conclusion

We studied how bees respond to an urban landscape in southern Brazil and found that plant specialists, ground-nesters, solitary bees, and kleptoparasites have a stronger response to urbanization than other groups. Banaszak-Cibicka and Dylewski (2021) recently argued that species loss to environmental changes could reflect losses of some functional groups, and total species richness may not be a good indicator in pollinator communities. We therefore conclude that special attention is necessary to conserve vulnerable groups rather than relying on the resilience of generalist eusocial species, such as some stingless bees. Among the predictors, vegetation cover proportion and landscape diversity had the greatest explanatory power, while distance among fragments and grassland cover were also relevant.

We suggest creating new green areas in the city and promoting vegetation heterogeneity in the existing ones by improving host plants and soil quality. Each country has different native species of flowers that attract bees, and their cultivation in gardens and parks should be promoted for bee conservation. In this context, programs that increase the bee-friendly

residential gardens are important, as well as the reduction of lawn mowing (Wastian et al. 2016; Bal-dock et al. 2019). Taken together, these actions help prevent local and regional extinctions.

Acknowledgements This study was funded by the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), who promoted a PhD scholarship to Leticia Vanessa Graf. We are grateful to Ministério do Meio Ambiente, Exército Brasileiro, Infraero and Secretaria do Meio Ambiente de Curitiba for research authorizations. Leticia Vanessa Graf received a PhD scholarship from Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq). Special thanks for Danilo Boscolo, Gabriel A. R. Melo and Isabela G. Varassin and for comments on this study and Eduardo A. B. de Almeida, Antonio J. C. Aguiar and Daniele R. Parizotto for functional traits' review.

Author contributions LVG and RBG conceived this research; LVG did the samplings; LVG and RBG identified the species; LVG and IS performed the analysis; LVG wrote and all authors participated in the revisions of it as well as in the interpretation of the data. All authors read and approved the final manuscript.

Funding This study was funded by the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), who promoted a PhD scholarship to Leticia Vanessa Graf.

Declarations

Conflict of interest The authors declare that they have no conflict of interest to declare that are relevant to the content of this article. The authors have no relevant financial or non-financial interests to disclose.

Ethical approval All procedures performed in this study were in accordance with the ethical standards and legislation of the institutions and sites where it was conducted.

References

- Aguirre-Gutiérrez J, Biesmeijer JC et al (2015) Susceptibility of pollinators to ongoing landscape changes depends on landscape history. *Divers Distrib* 21:1129–1140
- Aizen MA, Feinsinger P (2003) Bees not to be? Responses of insect pollinator faunas and flower pollination to habitat fragmentation. *Ecol Stud* 162:111–129
- Almeida EAB, Packer L, Melo GAR et al (2019) The diversification of neopasiphaeinae bees during the Cenozoic (Hymenoptera: Colletidae). *Zool Scr* 48:226–242
- Araújo ED, Costa M, Chaud-Netto J, Fowler HG (2004) Body size and flight distance in stingless bees (Hymenoptera: Meliponini): inference of flight range and possible ecological implications. *Braz J Biol* 64:563–568
- Ascher JS, Pickering J (2020) Discover life bee species guide and world checklist

- (Hymenoptera: Apoidea: Anthophila). http://www.discoverlife.org/mp/20q?guide=Apoidea_species. Accessed 20 Oct 2021
- Baldock KCR, Goddard MA, Hicks DM et al (2019) A systems approach reveals urban pollinator hotspots and conservation opportunities. *Nat Ecol Evol* 3:363–373
- Banaszak-Cibicka W, Dylewski L (2021) Species and functional diversity: a better understanding of the impact of urbanization on bee communities. *Sci Total Environ* 774:145729
- Banaszak-Cibicka W, Zmihorski M (2012) Wild bees along an urban gradient: Winners and losers. *J Insect Conserv* 16:331–343
- Banaszak-Cibicka W, Zmihorski M (2020) Are cities hotspots for bees? Local and regional diversity patterns lead to different conclusions. *Urban Ecosyst* 23:713–722
- Baselga A (2010) Partitioning the turnover and nestedness components of beta diversity. *Glob Ecol Biogeogr* 19:134–143
- Baselga A, Orme CDL (2012) Betapart: an R package for the study of beta diversity. *Methods Ecol Evol* 3:808–812
- Bauer DM, Sue Wing I (2016) The macroeconomic cost of catastrophic pollinator declines. *Ecol Econ* 126:1–13
- Biesmeijer JC, Roberts SPM, Reemer M et al (2006) Parallel declines in pollinators and insect-pollinated plants in Britain and the Netherlands. *Science* 313:351–354
- Boscolo D, Metzger JP (2009) Is bird incidence in Atlantic forest fragments influenced by landscape patterns at multiple scales? *Landsc Ecol* 24:907–918
- Boscolo D, Alves Ferreira P, Elsinor Lopes LE (2016) Da matriz à matiz: em busca de uma abordagem funcional na Ecologia de Paisagens. *Filos e História Da Biol* 11:157–187
- Boscolo D, Tokumoto PM, Ferreira PA et al (2017) Positive responses of flower visiting bees to landscape heterogeneity depend on functional connectivity levels. *Perspect Ecol Conserv* 15:18–24
- Buchholz S, Egerer MH (2020) Functional ecology of wild bees in cities: towards a better understanding of trait-urbanization relationships. *Biodivers Conserv* 29:2779–2801
- Burdine JD, McCluney KE (2019) Differential sensitivity of bees to urbanization-driven changes in body temperature and water content. *Sci Rep* 9:1–10
- Calvillo LM, Ramírez VM, Parra-Tabla V, Navarro J (2010) Bee diversity in a fragmented landscape of the Mexican neotropic. *J Insect Conserv* 14:323–334
- Cardinal S, Danforth BN (2013) Bees diversified in the age of Eudicots. *Proc R Soc B* 280:20122686
- Cardoso MC, Gonçalves RB (2018) Reduction by half: the impact on bees of 34 years of urbanization. *Urban Ecosyst* 21:943–949
- Cariveau DP, Winfree R (2015) Causes of variation in wild bee responses to anthropogenic drivers. *Curr Opin Insect Sci* 10:104–109. <https://doi.org/10.1016/j.cois.2015.05.004>
- Chapman RE, Bourke AFG (2001) The influence of sociality on the conservation biology of social insects. *Ecol Lett* 4:650–662
- Congedo L (2016) Semi-Automatic classification plugin documentation release 4.8.0.1. 201. <https://doi.org/10.13140/RG.2.2.29474.02242/1>
- Coutinho JG, Hipolito J, Santos RL, Moreira EF, Boscolo D, Viana BF (2021) Landscape structure is a major driver of bee functional diversity in crops. *Front Ecol and Evol*. <https://doi.org/10.3389/fevo.2021.624835>
- Danforth BN, Minckley RL, Neff JL (2019) The solitary bees: biology, evolution, conservation. Princeton University Press, Princeton
- Dearborn DC, Kark S (2010) Motivaciones para conservar la biodiversidad urbana. *Conserv Biol* 24:432–440
- Deguines N, Julliard R, de Flores M, Fontaine C (2016) Functional homogenization of flower visitor communities with urbanization. *Ecol Evol* 6:1967–1976
- Fahrig L (2003) Effects of habitat fragmentation on biodiversity. *Annu Rev Ecol Syst* 34:487–515
- Fahrig L, Baudry J, Brotons L et al (2011) Functional landscape heterogeneity and animal biodiversity in agricultural landscapes. *Ecol Lett* 14:101–112
- Ferreira PA, Boscolo D, Carvalheiro LG et al (2015) Responses of bees to habitat loss in fragmented landscapes of Brazilian Atlantic Rainforest. *Landsc Ecol* 30:2067–2078
- Ferreira PA, Boscolo D, Lopes LE et al (2020) Forest and connectivity loss simplify tropical pollination networks. *Oecologia* 192:577–590
- Flores LMA, Zanette LRS, Boscolo D et al (2019) Landscape structure effects on bee and wasp assemblages in a semi-arid buffer zone. *Landsc Online* 76:1–17
- Geslin B, Le Féon V, Folschweiller M et al (2016) The proportion of impervious surfaces at the landscape scale structures wild bee assemblages in a densely populated region. *Ecol Evol* 6:6599–6615
- Grime JP (1998) Benefits of plant diversity to ecosystems: immediate, filter and founder effects. *J Ecol* 86:902–910
- Hall DM, Camilo GR, Tonietto RK et al (2017) The city as a refuge for insect pollinators. *Conserv Biol* 31:24–29
- Harrison T, Gibbs J, Winfree R (2018) Forest bees are replaced in agricultural and urban landscapes by native species with different phenologies and life-history traits. *Glob Chang Biol* 24:287–296
- Hubbell SP (2001) The unified neutral theory of biodiversity and biogeography. Princeton University Press, Princeton
- IBGE (2020) <https://cidades.ibge.gov.br/brasil/pr/curitiba/panorama>. Accessed 08 Jan 2020
- Jackson HB, Fahrig L (2015) Are ecologists conducting research at the optimal scale? *Glob Ecol Biogeogr* 24:52–63
- Jaffé R, Castilla A, Pop N et al (2016) Landscape genetics of a tropical rescue pollinator. *Conserv Genet* 17:267–287
- Jones L, Leather R (2012) Invertebrates in urban areas: a review. *Eur J Entomol* 109:463–478
- Machado T, Viana BF, da Silva CI et al (2020) How landscape composition affects pollen collection by stingless bees? *Landsc Ecol* 35:747–759
- Martins AC, Gonçalves RB, Melo GAR (2013) Changes in wild bee fauna of a grassland in Brazil reveal negative effects associated with growing urbanization during the last 40 years. *Zoologia* 30:157–176
- Martins KT, Gonzalez A, Lechowicz MJ (2017) Patterns of pollinator turnover and increasing diversity associated with urban habitats. *Urban Ecosyst* 20:1359–1371

- Mazerolle MJ (2019) AICcmodavg: Model selection and multimodel inference based on (Q)AIC(c). R package version 2.2–2. <https://cran.r-project.org/package=AICcmodavg>
- McGarigal K, Cushman SA, Ene E (2012). FRAGSTATS v4: spatial pattern analysis program for categorical and continuous maps. In: Computer software program produced by the authors at the University of Massachusetts, Amherst.
- McKinney ML (2006) Urbanization as a major cause of biotic homogenization. *Biol Conserv* 127:247–260
- Melo AS (2008) O que ganhamos “confundindo” riqueza de espécies e equabilidade em um índice de diversidade? *Biota Neotrop* 8:21–27
- Melo GAR, Gonçalves RB (2005) Higher-level bee classifications (Hymenoptera, Apoidea, Apidae sensu lato). *Rev Bras Entomol* 22:153–159
- Michener CD (2007) The bees of the world, 2nd edn. The Johns Hopkins University Press, Baltimore
- Miguet P, Jackson HB, Jackson ND et al (2016) What determines the spatial extent of landscape effects on species? *Landsc Ecol* 31:1177–1194
- Moreira EF, Boscolo D, Viana BF (2015) Spatial heterogeneity regulates plant-pollinator networks across multiple landscape scales. *PLoS ONE* 10:1–19
- Nery LS, Takata JT, Camargo BB et al (2018) Bee diversity responses to forest and open areas in heterogeneous Atlantic Forest. *Sociobiology* 65:686–695
- Niebuhr BBS, Wosniack ME, Santos MC et al (2015) Survival in patchy landscapes: the interplay between dispersal, habitat loss and fragmentation. *Sci Rep* 5:1–10
- Oksanen J, Blanchet FG, Kindt R, et al (2013) Package ‘vegan’. <http://cran.r-project.org/web/packages/vegan/index.html>. Accessed 25 Dec 2019
- Olden JD, Rooney TP (2006) On defining and quantifying biotic homogenization. *Glob Ecol Biogeogr* 15:113–120
- Oliveira FPM, Absy ML, Miranda IS (2009) Recurso polínico coletado por abelhas sem ferrão (Apidae, Meliponinae) em um fragmento de floresta na região de Manaus - Amazonas. *Acta Amaz* 39:505–518
- Ollerton J, Winfree R, Tarrant S (2011) How many flowering plants are pollinated by animals? *Oikos* 120:321–326
- Orlandin E, Carneiro E (2021) Classes of protection in urban forest fragments are ineffective in structuring butterfly assemblages: landscape and forest structure are far better predictors. *Urban Ecosyst*. <https://doi.org/10.1007/s11252-020-01086-z>
- Packer L, Zayed A, Grixti JC et al (2005) Conservation genetics of potentially endangered mutualisms: reduced levels of genetic variation in specialist versus generalist bees. *Conserv Biol* 19:195–202
- Pereira FW, Carneiro L, Gonçalves RB (2020) More losses than gains in ground-nesting bees over 60 years of urbanization. *Urban Ecosyst*. <https://doi.org/10.1007/s11252-020-01030-1>
- Pielou EC (1966) The measurement of diversity in different types of biological collections. *J Theor Biol* 13:131–144
- QGIS Development Team (2019) QGIS Geographic Information System, Geospatial Foundation Project. <http://qgis.osgeo.org>
- R Development Core Team (2016) A language and environment for statistical computing, R Foundation for Statistical Computing. <http://www.R-project.org>
- Rocha-Filho LC, Ferreira-Caliman MJ, Garófalo CA, Augusto SC (2018) A specialist in an urban area: are cities suitable to harbour populations of the oligolectic bee *Centris (Melacentris) collaris* (Apidae: Centridini)? *Ann Zool Fennici* 55:135–149
- Rozen JG Jr (2001) A taxonomic key to mature larvae of cleptoparasitic bees (Hymenoptera: Apoidea). *Am Mus Nov* 3309:1–28
- Schneiberg I, Boscolo D, Devoto M et al (2020) Urbanization homogenizes the interactions of plant-frugivore bird networks. *Urban Ecosyst* 23:457
- Seto KC, Güneralp B, Hutyra LR (2012) Global forecasts of urban expansion to 2030 and direct impacts on biodiversity and carbon pools. *Proc Natl Acad Sci USA* 109:16083–16088
- Sheffield CS, Pindar A, Packer L et al (2013) The potential of cleptoparasitic bees as indicator taxa for assessing bee communities. *Apidologie* 44:501–510
- Silva DAT (2008) Caracterização da distribuição e riqueza de ninhos de Apini (Hymenoptera, Apidae) eussociais no contexto de um remanescente de Floresta Ombrófila Mista, Estação Experimental do Canguiri, Pinhais-PR. Dissertation, Universidade Federal do Paraná
- Silva DP, Nogueira DS, De Marco P (2017) Contrasting patterns in solitary and eusocial bees while responding to landscape features in the Brazilian cerrado: a multiscaled perspective. *Neotrop Entomol* 46:264–274
- Silveira FA, Melo GAR, Almeida EAB (2002) Abelhas brasileiras: Sistemática e Identificação
- Steffan-Dewenter I (2003) Importance of habitat area and landscape context for species richness of bees and wasps in fragmented orchard meadows. *Conserv Biol* 17:1036–1044. <https://doi.org/10.1046/j.1523-1739.2003.01575.x>
- Steffan-Dewenter I, Münzenberg U, Bürger C et al (2002) Scale-dependent effects of landscape context on three pollinator guilds. *Ecology* 83:1421–1432
- Taura HM, Laroca S (2001) A associação de abelhas silvestres de um biótopo urbano de Curitiba (Brasil), com comparações espaço-temporais: abundância relativa, fenologia, diversidade e exploração de recursos (Hymenoptera, Apoidea). *Acta Biológica Parana* 30:35–137
- Threlfall CG, Walker K, Williams NSG et al (2015) The conservation value of urban green space habitats for Australian native bee communities. *Biol Conserv* 187:240–248
- Tylianakis JM, Tscharnkte T, Lewis OT (2007) Habitat modification alters the structure of tropical host-parasitoid food webs. *Nature* 445:202–205
- United Nations (2017) <https://www.un.org/development/desa/en/news/population/world-population-prospects-2017.html>. Accessed 08 Jan 2020
- UNRIC (2018) <https://www.unric.org/pt/actualidade/31537>. Accessed 02 July 2020
- Vanbergen AJ, Garratt MP, Vanbergen AJ et al (2013) Threats to an ecosystem service: pressures on pollinators. *Front Ecol Environ* 11:251–259
- Wastian L, Unterweger PA, Betz O (2016) Influence of the reduction of urban lawn mowing on wild bee diversity (Hymenoptera, Apoidea). *J Hymenopt Res* 49:51–63
- Wenzel A, Grass I, Belavadi VV, Tscharnkte T (2020) How urbanization is driving pollinator diversity and pollination: a systematic review. *Biol Conserv* 241:108321

- Williams NM, Crone EE, Roulston TH et al (2010) Ecological and life-history traits predict bee species responses to environmental disturbances. *Biol Conserv* 143:2280–2291
- Wilson CJ, Jamieson MA (2019) The effects of urbanization on bee communities depends on floral resource availability and bee functional traits. *PLoS ONE* 14:1–18
- Winfree R, Bartomeus I, Cariveau DP (2011) Native pollinators in anthropogenic habitats. *Annu Rev Ecol Evol Syst* 42:22
- Wright IR, Roberts SPM, Collins BE (2015) Evidence of forage distance limitations for small bees (Hymenoptera: Apidae). *Eur J Entomol* 112:303–310
- Xie Z, Qiu J, Chen X (2013) Decline of nest site availability and nest density of underground bees along a distance gradient from human settlements. *Entomol Sci* 16:170–178
- Zanette LRS, Martins RP, Ribeiro SP (2005) Effects of urbanization on Neotropical wasp and bee assemblages in a Brazilian metropolis. *Landsc Urban Plan* 71:105–121

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.